

## Part 2: Applying OLTP to OLAP Transformation

### Task 1: MySQL OLTP Schema

File Name: Part 2\_Task 1.sql

SQL

-- Patients Table

```
CREATE TABLE Patients (  
    PatientID INT AUTO_INCREMENT PRIMARY KEY,  
    FirstName VARCHAR(50) NOT NULL,  
    LastName VARCHAR(50) NOT NULL,  
    DOB DATE NOT NULL,  
    Gender VARCHAR(10),  
    ContactNumber VARCHAR(15)  
);
```

-- Doctors Table

```
CREATE TABLE Doctors (  
    DoctorID INT AUTO_INCREMENT PRIMARY KEY,  
    FirstName VARCHAR(50) NOT NULL,  
    LastName VARCHAR(50) NOT NULL,  
    Specialization VARCHAR(100)  
);
```

-- Appointments Table

```
CREATE TABLE Appointments (  
    AppointmentID INT AUTO_INCREMENT PRIMARY KEY,  
    PatientID INT,  
    DoctorID INT,  
    AppointmentDate TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    Status VARCHAR(20) CHECK (Status IN ('Scheduled', 'Completed', 'Cancelled')),  
    FOREIGN KEY (PatientID) REFERENCES Patients(PatientID) ON DELETE CASCADE,  
    FOREIGN KEY (DoctorID) REFERENCES Doctors(DoctorID) ON DELETE CASCADE  
);
```

-- Medical Records Table

```
CREATE TABLE MedicalRecords (  
    RecordID INT AUTO_INCREMENT PRIMARY KEY,  
    PatientID INT,  
    DoctorID INT,  
    Diagnosis TEXT,  
    Prescription TEXT,  
    VisitDate TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    FOREIGN KEY (PatientID) REFERENCES Patients(PatientID) ON DELETE CASCADE,  
    FOREIGN KEY (DoctorID) REFERENCES Doctors(DoctorID) ON DELETE CASCADE
```

```
);
```

```
-- Billing Table
```

```
CREATE TABLE Billing (  
    BillID INT AUTO_INCREMENT PRIMARY KEY,  
    PatientID INT,  
    Amount DECIMAL(10,2) NOT NULL,  
    PaymentStatus VARCHAR(20) CHECK (PaymentStatus IN ('Paid', 'Pending', 'Overdue')),  
    FOREIGN KEY (PatientID) REFERENCES Patients(PatientID) ON DELETE CASCADE  
);
```

```
-- INSERT Script
```

```
INSERT INTO Patients (FirstName, LastName, DOB, Gender) VALUES ('John', 'Doe', '1985-01-01',  
'Male');
```

```
INSERT INTO Doctors (FirstName, LastName, Specialization) VALUES ('Jane', 'Smith', 'Cardiology');
```

```
INSERT INTO Appointments (PatientID, DoctorID, Status) VALUES (1, 1, 'Completed');
```

```
INSERT INTO MedicalRecords (PatientID, DoctorID, Diagnosis, Prescription) VALUES (1, 1,  
'Hypertension', 'Lisinopril');
```

```
INSERT INTO Billing (PatientID, Amount, PaymentStatus) VALUES (1, 150.00, 'Paid');
```